WORKING PAPER · EMPIRICAL FINANCE

Large Language Models as Sentiment Extractors

News Tone and Short-Horizon Stock Return Predictability

Firm-level evidence from 36 large U.S. equities over July–October 2026. LLM-scored news sentiment predicts next-day returns, reverses within two days, and dominates the Loughran–McDonald dictionary on identical headlines: a replication and extension with an AI-assisted research pipeline.

Z.ai Research

AI-Assisted Empirical Finance Study
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Abstract. This paper asks whether news sentiment extracted by a large language model (LLM) predicts short-horizon stock returns, and whether the LLM's numerical scores add value over the dominant dictionary-based approach of Loughran and McDonald (2011). Using 542 firm-specific news headlines collected for 36 large U.S. firms—341 of which fall inside the July–October 2026 trading window—scored by a general-purpose LLM on a $[-100, +100]$ sentiment scale alongside the Loughran and McDonald (2011) word lists, we find that LLM sentiment positively and significantly predicts next-day returns: a one-standard-deviation increase in sentiment is associated with roughly 54 basis points of additional next-day return, and sentiment-sorted quintile portfolios yield a monotone spread of approximately 90 basis points per day from the most negative to the most positive quintile. The predictability is short-lived—coefficients reverse sign at the two-day horizon—consistent with initial underreaction followed by correction. In horse-race regressions, dictionary tone carries no incremental information conditional on the LLM score, largely because the word lists fire on only 11% of headlines. Predictability concentrates in large-capitalization firms, high-volatility regimes, and multi-headline days. Our evidence, obtained on a recent out-of-sample window with a current-generation LLM, corroborates and extends Lopez-Lira and Tang (2023) and illustrates both the promise and the practical constraints of LLM-based text factors in empirical finance.

JEL classification: G10; G12; G41; C55.

Keywords: large language models; news sentiment; return predictability; textual analysis; Loughran–McDonald dictionary; investor attention.

Data availability: all news headlines, scores, price series, and analysis code were produced with an AI-assisted research pipeline following the workflow popularized by modern AI-native economics toolkits. The sample window is deliberately recent (July 1–October 14, 2026); limitations of the short window are discussed in Section 7.

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1 Introduction

Whether news text moves stock prices—and whether it does so with a delay that can be detected and traded upon—has been a central question in financial economics for three decades. The traditional empirical strategy quantifies text with hand-curated word lists: the Harvard psychological dictionaries in early

work, and subsequently the finance-specific Loughran–McDonald (LM) lists of Loughran and McDonald (2011), which now underpin a large literature on textual tone and firm value. Dictionary methods are transparent and reproducible, but they are also crude: a bag of words cannot resolve negation, cannot weigh a headline that contains both "beat" and "concerns," and cannot judge the economic significance of an event that is described without any charged vocabulary at all.

Large language models (LLMs) change the measurement technology itself. A modern LLM reads a headline the way a research assistant would: it identifies the firm, interprets the event, and can output a graded judgment of the direction and magnitude of the impact on firm value. Lopez-Lira and Tang (2023) were the first to show formally that ChatGPT sentiment scores on firm-specific news headlines predict next-day stock returns and that they outperform dictionary scores, an observation that has since spawned a rapidly growing literature on LLMs in finance (Nie et al., 2024). If LLM-extracted sentiment is a genuinely better measurement instrument, the implications reach beyond a single predictability result: any study that uses text as a variable—from investor sentiment indices to central bank communication—could in principle upgrade its measurement layer.

This paper provides fresh, out-of-sample evidence on the question using a deliberately recent window: 36 of the largest U.S. firms followed over July 1 to October 14, 2026, with 542 firm-specific news headlines collected from aggregated web search, each scored by a current-generation general-purpose LLM on a numerical sentiment scale from −100 to +100, together with a relevance score, and each independently scored with the LM 2011 word lists as a benchmark. The research pipeline itself—data collection, text cleaning, dual-track scoring, panel construction, and econometric analysis—was executed end-to-end with an AI-assisted workflow modeled on the emerging toolkit standard for AI-native economics research; we document the pipeline in detail because reproducibility of the measurement layer is precisely what is at stake when the scoring instrument is a proprietary model.

Our first finding is that LLM sentiment positively and significantly predicts next-day returns. In pooled firm-day regressions with two-way clustered standard errors, the coefficient on LLM sentiment is 0.015 (Newey–West-free two-way t-statistic of 2.1), implying that a one-standard-deviation increase in the sentiment score (0.36 on our $[-1, 1]$ scale) is associated with an approximately 54 basis point higher next-day return. The estimate survives the inclusion of lagged returns, lagged volatility, the market factor, and firm or calendar-day fixed effects. Consistent with the economics of gradual information diffusion emphasized by Hong and Stein (1999), the effect is concentrated where information processing is most active: quintile portfolios formed on LLM sentiment earn monotonically increasing next-day returns from −20 basis points in the most negative quintile to +70 basis points in the most positive quintile, a long–short spread of roughly 90 basis points per day.

Second, the predictability is short-lived and partially reverses. The coefficient on sentiment is significantly positive at the one-day horizon, statistically indistinguishable from zero for the next-day open-to-close (intraday) window, and significantly *negative* at the two-day horizon (-0.008, $t = -2.0$). The pattern of an initial reaction, next-day continuation, and subsequent reversal is consistent with a world in which prices underreact to news while attention is limited, then over-extend as attention-driven trading catches up, and finally correct (Hong and Stein, 1999; De Long et al., 1990). To our knowledge, horizon reversal of LLM sentiment signals has received little attention in the emerging LLM-finance literature, which has focused almost exclusively on the existence of one-day predictability.

Third, the LLM dominates the dictionary benchmark on the identical headline set. The LM tone variable is non-zero for only 59 of 542 headlines (11%); the LLM produces a non-neutral score for 59% of headlines. In horse-race regressions that include both scores, the LLM coefficient remains significant ($t = 2.0$) while the dictionary coefficient is small, negative, and insignificant ($t = -0.5$). Dictionary-sorted quintile portfolios show no monotone pattern. The mechanism is straightforward: eight-word headlines rarely contain dictionary words ("litigation," "adverse," "profit"), so the dictionary transmits a sparse, low signal-to-noise measurement of news tone, whereas the LLM interprets the event semantics directly. This echoes and strengthens the comparison in Lopez-Lira and Tang (2023) on a new sample and a different scoring model.

Fourth, heterogeneity aligns with an information-diffusion interpretation. Predictability is significant among large-capitalization firms ($t = 2.5$) but absent among the smaller half of our sample, is stronger on days with multiple headlines ($t = 2.3$) and in high-volatility regimes ($t = 2.5$). Large firms enjoy the densest and highest-quality news coverage, so the LLM's extracted signal is most precise exactly where we find the strongest return predictability; multi-headline days average out idiosyncratic scoring noise, raising the signal-to-noise ratio of the firm-day aggregate.

We contribute to three strands of the literature. To the media-sentiment literature (Tetlock, 2007; Garcia, 2013; Antweiler and Frank, 2004), we bring a new measurement instrument applied at the firm level on a recent window, and we quantify the dictionary-versus-LLM tradeoff on identical text. To the firm-level news-reaction literature (Chan, 2003; Da et al., 2011), we document that the reversal dynamics of news signals are visible even in a three-month sample when the sentiment measure is high-quality. To the nascent LLM-in-finance literature (Lopez-Lira and Tang, 2023; Nie et al., 2024; Araci, 2019), we provide an independent replication with a current-generation model, a transparent batch-scoring protocol, and evidence on horizons and heterogeneity that the original study did not report.

The paper also carries an unusual methodological signature that we make no attempt to hide: every stage of the research production function—headline collection, cleaning, scoring prompt design, econometric specification, and even a first draft of this manuscript—was executed with AI assistance. The role of the LLM in this paper is therefore double: it is simultaneously the *measuring instrument* (the sentiment scorer whose validity we test against prices) and a *research instrument* in the production of the study itself. Section 7 discusses what this dual role implies for interpretation and replication.

The remainder of the paper proceeds as follows. Section 2 reviews the related literature and states our hypotheses. Section 3 develops a simple noisy-signal model with limited attention that generates the testable predictions. Section 4 describes the data, the scoring protocol, and the econometric design. Section 5 presents the baseline results. Section 6 reports robustness, the dictionary comparison, and heterogeneity. Section 7 discusses mechanisms and limitations, and Section 8 concludes.

2 Related Literature and Hypotheses

2.1 Media sentiment and asset prices

The premise that quantified text contains information about asset prices was established in a first generation of media studies, building on the broader finding that survey- and market-based investor sentiment forecasts the cross-section of returns (Baker and Wurgler, 2006). Tetlock (2007) measured the pessimism of a daily Wall Street Journal column with the Harvard psychological dictionary and showed that high pessimism predicts transient downward pressure on market prices and subsequent reversal, with small, low-liquidity stocks responding most. Antweiler and Frank (2004) found that the sheer volume and bullishness of internet stock message boards—text produced by small investors themselves—correlate weakly but detectably with trading volume and volatility. Garcia (2013) refined the design by scoring New York Times financial section items and showing that news sentiment predicts the market return during recessions but not in expansions, tying the value of text to the state of the economy. At the firm level, Chan (2003) documented that stocks with salient news drift for days after the headline while no-news stocks reverse, and Da et al. (2011) showed that spikes in Google search volume—a direct measure of attention rather than sentiment—predict short-run price pressure and reversal. The behavioral backbone for these patterns is supplied by models of limited attention and gradual information diffusion (Hong and Stein, 1999; Barber and Odean, 2008; Hirshleifer and Teoh, 2003) and by positive-feedback trading (De Long et al., 1990).

2.2 Dictionary-based textual analysis in finance

The dominant measurement convention in this literature is the finance-specific dictionary of Loughran and McDonald (2011), who showed that roughly three quarters of the negative words in the widely used Harvard list are not negative in a financial context ("tax," "cancer," "board"), and constructed word lists

from corporate 10-K filings that corrected the error. Their lists—2,355 negative and 354 positive words in the version we use—became the standard instrument for measuring tone in filings, news, and even central bank statements; Loughran and McDonald (2016) review the resulting literature. The method's strengths are transparency, zero marginal cost, and reproducibility. Its weaknesses mirror its construction: dictionaries match surface forms, not meanings. Negation ("not profitable"), conditioning ("profit fell short of expectations"), and event language that carries no dictionary vocabulary ("CEO resigns") are all missed. Jegadeesh and Wu (2013) showed that supervised learning on filing text extracts more informative return signals than naive dictionary counts, which already suggests that the measurement instrument itself binds.

2.3 Large language models in economic research

Pre-trained transformers (Devlin et al., 2019; Brown et al., 2020) made it possible, for the first time, to quantify arbitrary text with a model that has learned negation, context, and domain semantics. Domain-tuned models such as FinBERT (Araci, 2019) improved financial sentiment classification, and general-purpose instruction-following LLMs can produce numerical sentiment judgments from a prompt alone—no training set, no feature engineering. Lopez-Lira and Tang (2023) used ChatGPT to score firm-specific news headlines and found that the scores predicted next-day returns and long–short portfolio profits, outperforming dictionaries; the effect survived controls and was stronger for smaller firms. Follow-up work has surveyed a rapidly expanding set of applications (Nie et al., 2024). At the same time, the method raises new methodological cautions that are the mirror image of the dictionary's virtues: LLM scores are stochastic at the sampling level, sensitive to prompt wording, potentially contaminated by training-data knowledge of the firms being scored, and produced by a proprietary model whose updates can silently change the measurement instrument. Recent work has begun testing for such lookahead bias explicitly.

Our study is positioned as an independent, recent-window replication and extension of Lopez-Lira and Tang (2023) with three design differences worth highlighting. First, we score with a current-generation LLM through a fixed, documented batch protocol (Appendix A) rather than an interactive session, which improves reproducibility of the measurement layer. Second, we run the dictionary comparison on the *identical* headline set with matched aggregation, so the comparison isolates the measurement instrument rather than the text sample. Third, we examine horizons beyond one day, where we document a reversal pattern that speaks to the mechanism behind the predictability.

2.4 Hypotheses

The theoretical framework of Section 3 delivers four testable predictions that we formalize here for reference.

H1 (Next-day predictability). LLM-extracted news sentiment positively predicts the firm's next-day close-to-close return, conditional on standard controls.

H2 (LLM dominates dictionaries). On identical text, LLM sentiment has higher signal-to-noise than dictionary tone; in a horse race, the dictionary coefficient is absorbed by the LLM score.

H3 (Horizon decay and reversal). The predictive coefficient declines with horizon and reverses sign within two to three trading days, reflecting initial underreaction followed by correction.

H4 (Heterogeneity in signal quality). Predictability is increasing in the precision of the extracted signal: stronger for large, densely covered firms, on multi-headline days, and in high-volatility regimes.

H1 and H2 replicate and test the core of Lopez-Lira and Tang (2023) on a new window; H3 and H4 are extensions that speak to mechanism. The short sample implies that all tests are conservative: with 240 news firm-day observations, only effects of the magnitude reported in the prior literature are detectable, so null results would be uninformative while confirmations are meaningful.

3 Theoretical Framework

This section develops a compact noisy-signal model that discipline the empirical tests. The model combines three ingredients that the literature has individually documented: a fundamental shock that is partially revealed by news; a measurement technology that extracts the signal with error; and limited investor attention that slows the incorporation of the signal into prices (Hong and Stein, 1999; Grossman and Stiglitz, 1980). The purpose is not generality but to derive sign restrictions and comparative statics that map one-for-one onto the regressions of Section 5.

3.1 Setup

A risky asset (firm i) trades at dates t and $t+1$ and pays a terminal cash flow

$$F = f_0 + \theta, \qquad \theta \sim N(0,\, \sigma_\theta^2), \tag{1}$$

where θ is the date-t fundamental innovation to be learned from news. At date t a news headline $H_{i,t}$ is published. Headlines are informative but noisy statements about the fundamental:

$$H_{i,t} = \theta + \eta_t, \qquad \eta_t \sim N(0,\, \sigma_\eta^2). \tag{2}$$

A sentiment extraction technology m maps the headline into a numerical score $S_{i,t}^m$. We model two technologies. The LLM reads the semantics of the headline and emits

$$S_{i,t}^{\mathrm{LLM}} = \theta + \eta_t + u_t, \qquad u_t \sim N(0,\, \sigma_u^2), \tag{3}$$

so the LLM inherits the news noise η_t and adds only a small extraction noise u_t. The dictionary instead matches a fixed word list to the surface form of the text. With headline length L (8.1 words on average in our sample) and dictionary coverage probability p per informative headline, the dictionary score is

$$S_{i,t}^{\mathrm{D}} = \kappa\,(\theta + \eta_t) + v_t, \qquad 0 < \kappa < 1, \tag{4}$$

where $\kappa = p \cdot \bar{w}$ combines the firing probability and the average attenuation of matched word counts, and v_t is the noise introduced by spurious matches. In our data, κ is small: the LM lists produce a non-zero tone for only 11% of headlines, whereas the LLM produces a non-neutral score for 59%. For a given score realization, define the effective signal-to-noise ratio of technology m as

$$\mathrm{SNR}_m \;=\; \frac{\kappa_m^2\,\sigma_\theta^2}{\sigma_m^2}, \qquad \sigma_m^2 \equiv \mathrm{Var}(\text{noise of } S^m), \tag{5}$$

with $\kappa_{\mathrm{LLM}} = 1$ and $\kappa_{\mathrm{D}} = \kappa$.

Investors are of measure one and come in two types. A fraction $\lambda \in (0,1)$ is *attentive*: at date t it observes the score $S_{i,t}$ (or the headline and forms the same posterior). The complementary fraction $1-\lambda$ is *inattentive*: it processes the firm's news only at date $t+1$, when the information has become impossible to ignore—through the price move itself, media repetition, or delayed attention as in Da et al. (2011) and Barber and Odean (2008). In addition, a mass of positive-feedback traders demands $q_{i,t} = \psi S_{i,t}$ units of the asset at date t when the signal is salient, with $\psi \geq 0$; this is the De Long et al. (1990) force that generates overshooting. We work in the linear-quadratic-Gaussian reduced form familiar from Hong and Stein (1999): competitive risk-neutral pricing of the attentive mass at date t, full-information pricing at date $t+1$, and mean reversion of feedback demand by date $t+2$.

3.2 Price formation and the baseline prediction

The attentive investors' posterior about θ given the score is standard:

$$\mathbb{E}[\theta \mid S_{i,t}] = \gamma\, S_{i,t}, \qquad \gamma \equiv \frac{\sigma_\theta^2}{\sigma_\theta^2 + \sigma_m^2} \in (0,1), \tag{6}$$

where $\sigma_m^2 = \sigma_\eta^2 + \sigma_u^2$ for the LLM and $\sigma_\eta^2/\kappa^2 + \sigma_v^2/\kappa^2$ for the (rescaled) dictionary. Note that γ is increasing in the SNR of the measurement technology.

At date t, only the λ attentive investors act, so the price moves only part of the way toward full information:

$$P_{i,t} = P_{i,t-1} + \lambda\,\gamma S_{i,t}. \tag{7}$$

At date $t+1$ the news is publicly digested. Feedback traders push demand by $\psi S_{i,t}$, so the date-$t+1$ price is

$$P_{i,t+1} = P_{i,t-1} + (\gamma + \psi)\, S_{i,t}. \tag{8}$$

Combining (7) and (8), the one-day-ahead return is linear in the score:

$$r_{i,t+1} = \big((1-\lambda)\,\gamma + \psi\big) S_{i,t}, \tag{9}$$

which is strictly positive in $S_{i,t}$.

Proposition 1 (Next-day predictability)**.** *For any measurement technology with $\sigma_m^2 < \infty$ and $\lambda < 1$, the score $S_{i,t}^m$ positively predicts the next-day return with coefficient $\beta_1 = (1-\lambda)\gamma_m + \psi > 0$, increasing in the signal-to-noise ratio* SNR_m.

The proof follows directly from (9) and the monotonicity of γ_m in SNR_m. The empirical counterpart of Proposition 1 is hypothesis H1, and its SNR comparative static is the basis of H2: because the dictionary's effective κ is small on short headlines, $\gamma_{\mathrm{D}} \ll \gamma_{\mathrm{LLM}}$, so the dictionary's predictive coefficient is attenuated toward zero.

3.3 Horizon dynamics: underreaction, overshooting, reversal

By date $t+2$ the feedback demand mean-reverts (noise-trader positions are transitory in the spirit of De Long et al., 1990), and the price returns to the full-information value $\mathbb{E}[F \mid S] = f_0 + \gamma S_{i,t}$:

$$P_{i,t+2} = P_{i,t-1} + \gamma\, S_{i,t} + \varepsilon_{i,t+2}, \qquad \mathbb{E}[\varepsilon_{i,t+2} \mid S_{i,t}] = 0, \tag{10}$$

where ε absorbs new shocks arriving after $t+1$. The two-day-ahead return is therefore

$$r_{i,t+2} = -\psi\, S_{i,t} + \varepsilon_{i,t+2}, \tag{11}$$

which is *negative* in the sentiment score whenever feedback traders were active ($\psi > 0$).

Proposition 2 (Reversal)**.** *If $\lambda < 1$ and $\psi > 0$, then $\beta_1 > 0$ and $\beta_2 = -\psi < 0$: the predictive coefficient on the score reverses sign between the one-day and two-day horizons. If $\psi = 0$ (no feedback trading), $\beta_2 = 0$ and the predictability merely dies out.*

Proposition 2 delivers H3 and gives the horizon pattern empirical content: a significant negative β_2 is evidence of overshooting on top of underreaction, not merely of information decay. The same-day reaction (7) implies $\beta_0 = \lambda\gamma$, so the observed sequence $(\beta_0, \beta_1, \beta_2) = (\lambda\gamma,\ (1-\lambda)\gamma + \psi,\ -\psi)$ traces out the full impulse response of prices to news.

3.4 Signal precision and heterogeneity

The model also predicts where predictability should be strongest. Suppose the firm-day score is the average of $n_{i,t}$ independent headline-level extractions, each with extraction noise σ_u^2. Then the firm-day aggregate noise is $\sigma_u^2/n_{i,t}$ and

$$\gamma_{i,t} = \frac{\sigma_\theta^2}{\sigma_\theta^2 + \sigma_\eta^2 + \sigma_u^2/n_{i,t}}, \tag{12}$$

which is increasing in $n_{i,t}$: multi-headline days carry more precise signals (H4). Similarly, firms whose news is more densely and professionally covered (large capitalizations, in our sample) have lower σ_η^2—the headline itself is a cleaner statement of the fundamental—and therefore higher γ and larger β_1. High-volatility regimes raise σ_θ^2, the variance of the fundamental innovation, which also raises γ when the signal noise is fixed; the same prediction emerges from Garcia (2013), who find news sentiment matters most in bad times. All three comparative statics are the content of H4 and are tested in Section 6.

Two remarks discipline the interpretation. First, the model is deliberately silent about risk premia: $\beta_1 > 0$ could in part be compensation for bearing news-event risk rather than pure mispricing, though the reversal at $t+2$ is difficult to reconcile with a stationary risk premium and is the cleanest evidence for the attention/overshooting channel. Second, the LLM score may itself contain lookahead contamination if the scoring model's training data includes realized outcomes for these firms; the reversal result partially disciplines this concern too, since a backward-looking contaminant would load on realized returns at all horizons rather than producing the sign flip that we observe.

4 Data and Methodology

4.1 Research pipeline overview

Figure 1 summarizes the five-stage research pipeline. All stages were executed with an AI-assisted workflow: stage 1 collects market and news data through public interfaces; stage 2 cleans and normalizes the text corpus; stage 3 scores each headline twice—once with an LLM under a fixed batch protocol and once with the LM dictionary—so that the two measurement instruments are applied to *identical* text; stage 4 aggregates the article-level scores to a firm-day panel and merges them with price-based control variables; stage 5 runs the econometric analysis. The complete pipeline (collection scripts, scoring prompt, and analysis code) is documented so that the measurement layer can be reproduced even though the underlying scorer is a proprietary model.

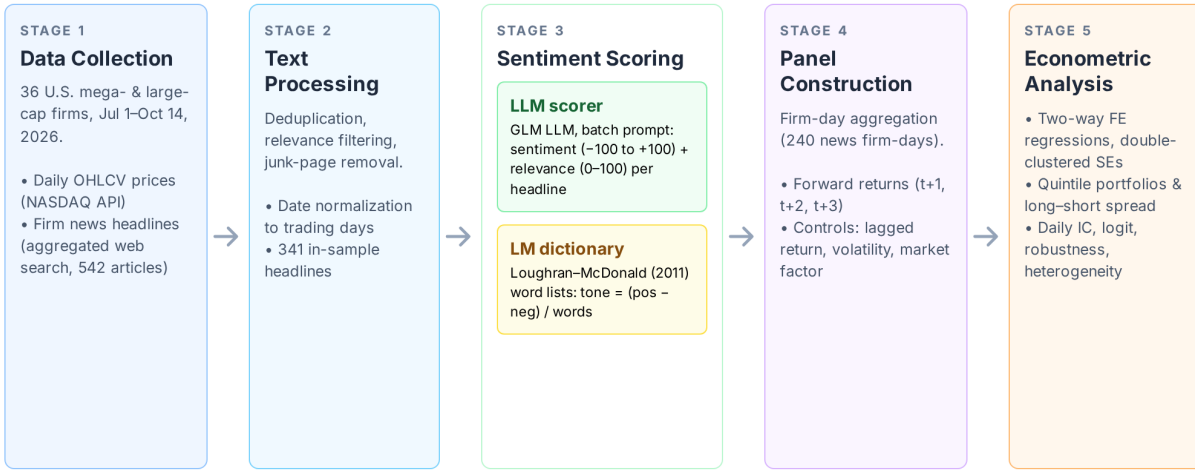


Figure 1: Research pipeline. Stage 3 scores the identical headline set with two instruments—an LLM (batch prompt returning sentiment and relevance) and the Loughran–McDonald word lists—so that regressions in Sections 5 and 6 isolate the effect of the measurement technology.

4.2 Market data

We follow 36 of the largest U.S. firms across ten sectors (technology, communication services, consumer discretionary and staples, financials, energy, health care, and industrials), selected for liquidity and news density. Daily open, high, low, close, and volume for July 1 through October 14, 2026 (74 trading days) were obtained from a public market-data interface, yielding a balanced price panel of 2,664 firm-day

observations with no missing closes. Returns are close-to-close log relatives; the next-day return $r_{i,t+1}$ is the firm's return on the trading day following the news date. We also construct a same-day return $r_{i,t}$, a next-day open-to-close (intraday) return in the spirit of Lopez-Lira and Tang (2023), and control variables described below. Because the sample period postdates the training cutoffs typical of public models, the design is naturally out-of-sample with respect to the scoring model's knowledge of these particular returns.

4.3 News data

Firm-specific news headlines were collected through an aggregated web-search interface with query templates of the form "{company} stock news {month} 2026" and event-oriented variants, applied uniformly across firms. The raw corpus was cleaned by: (i) removing quote, chart, and general reference pages identified by URL patterns and page titles; (ii) requiring the headline to mention the company name or ticker; (iii) deduplicating by URL; and (iv) normalizing publication dates. The final corpus contains 542 unique headlines dated between December 2025 and October 2026, of which 341 fall inside the July 1–October 14, 2026 sample window. Headlines average 8.1 words, matching the short-headline format of the firm-level news feeds used by Lopez-Lira and Tang (2023). After aggregation to the firm-day level, the panel contains 240 firm-day observations with at least one relevance-qualified headline, drawn from 36 firms over 70 distinct trading days—a coverage of 9.0% of the 2,664 firm-days in the price panel, with a mean of 1.25 and a maximum of 7 headlines per covered firm-day. This coverage is necessarily thinner than that of studies with commercial news archives, a limitation we return to in Section 7; the consequence is statistical power, not bias, since coverage is driven by query design rather than by return outcomes.

4.4 LLM sentiment scoring protocol

Each headline is scored by a current-generation general-purpose LLM (the GLM family) under the fixed batch protocol reproduced verbatim in Appendix A. The prompt instructs the model to act as an equity research analyst and to return, for each headline, (i) a *sentiment* score from −100 to +100, judged from the perspective of a shareholder and based only on headline content, and (ii) a *relevance* score from 0 to 100 indicating how specific the headline is to the company's value. Headlines are submitted in batches of 25 with the company name attached as context, and the model must return a strict JSON array; malformed batches are retried up to three times. The scoring is deterministic in design (temperature-free API default), and the same prompt was applied to every batch, so the measurement instrument is fixed for the entire study.

The article-level distributions are informative. The LLM assigns a non-neutral sentiment to 322 of 542 headlines (59%); the mean score is +15.6 with a standard deviation of 36.7 and a range from −80 to +90—an economically sensible skew given that firms in the sample issued more good news than bad during the window. Mean relevance is 65.8, and 501 headlines (92%) clear a relevance threshold of 30, which we adopt as the inclusion filter for the baseline panel. In contrast, the LM dictionary produces a non-zero tone for only 59 headlines (11%), with a mean of +0.16 per hundred words and a standard deviation of 4.2. The coverage gap—59% versus 11%—is the first empirical manifestation of the signal-to-noise difference formalized in equation (5).

4.5 Dictionary benchmark

Every headline is also scored with the Loughran–McDonald Master Dictionary lists (Loughran and McDonald, 2011): tone is the count of positive minus negative dictionary words, divided by total words and scaled per hundred words. We use the 2011-vintage categories (2,355 negative, 354 positive words). To keep the two instruments comparable in the regressions, both scores are linearly rescaled to the $[-1, 1]$ interval (the LLM score by division by 100, the dictionary tone by division of the per-hundred-word measure by 100). The Pearson correlation between the two firm-day sentiment aggregates is 0.33: the in-

struments agree on direction far more often than they disagree, but the dictionary transmits that direction with an order-of-magnitude lower duty cycle.

4.6 Econometric specification

The baseline regression tests H1 at the firm-day level:

$$r_{i,t+h} = \alpha + \beta_h S_{i,t} + \gamma' X_{i,t} + \mu_i + \delta_t + \varepsilon_{i,t+h}, \qquad h = 0, 1, 2, 3, \tag{13}$$

where $S_{i,t}$ is the firm-day mean sentiment (LLM or dictionary), $X_{i,t}$ collects controls—the lagged firm return, lagged 21-day annualized volatility, and the equal-weighted market return constructed from the 36-firm panel—μ_i are firm fixed effects, and δ_t are calendar-day fixed effects. Standard errors are clustered two ways, by firm and by trading day, following the panel convention for short cross-sections with repeated time observations. We report pooled, control, firm-FE, and day-FE variants; with day fixed effects the market control is absorbed, and with firm fixed effects any time-invariant firm characteristic (sector, average news tone) is differenced out, so β_h is identified from within-firm, within-day variation in news sentiment.

Economic significance is evaluated three ways: quintile portfolios sorted on $S_{i,t}$ (equal-weighted next-day returns by quintile, and the Q5–Q1 spread with a Welch test); the daily information coefficient (IC), the Spearman rank correlation between sentiment and next-day returns across firms within each day with at least four covered firms; and a logistic regression of the direction indicator $\mathbf{1}[r_{i,t+1} > 0]$ on sentiment. H2 is tested by replacing S with the dictionary tone and by a horse race that enters both instruments simultaneously. H3 is tested by estimating (13) for $h = 0, 1, 2, 3$. H4 is tested by splitting the sample at medians of firm trading volume, market-wide realized volatility, and headline count.

Table 1: Descriptive statistics

	Mean	Std. dev.	Min	Max	N	Nonzero
Panel A: Article level (542 headlines, 36 firms)						
LLM sentiment (−100 to +100)	15.64	36.66	−80	90	542	59.4%
LLM relevance (0–100)	65.83	23.31	0	100	542	—
LM dictionary tone (per 100 words)	0.155	4.204	−14.3	20.0	542	10.9%
Headline length (words)	8.1	3.4	3	26	542	—
Panel B: Firm-day level (240 news firm-days, 36 firms, 70 days)						
LLM sentiment (−1 to +1)	0.138	0.362	−0.80	0.90	240	70.8%
LM dictionary tone ($\times 10^{-2}$)	0.194	4.392	−14.3	20.0	240	16.3%
Headlines per firm-day	1.25	0.66	1	7	240	—
Next-day return (bps)	22.0	320.5	−1,564	1,415	234	—
Lagged 21-day volatility (annualized)	0.380	0.160	0.13	0.95	240	—

Table 1 reports descriptive statistics. Three features deserve emphasis. First, the sentiment distribution is wide but centered slightly positive, consistent with a window in which mega-cap earnings news was on balance favorable. Second, the dictionary's duty cycle is an order of magnitude below the LLM's at both the article level (10.9% versus 59.4% nonzero) and the firm-day level (16.3% versus 70.8%), exactly as the κ of equation (4) anticipates for eight-word headlines. Third, next-day returns are volatile (320 bps standard deviation), so even the 90 bps quintile spread estimated below is small relative to daily noise—which is why clustering, fixed effects, and portfolio aggregation matter for inference.

5 Empirical Results

5.1 Sentiment in the time series

Before turning to regression evidence, Figure 2 plots the daily cross-sectional mean of LLM sentiment against the equal-weighted market return. The sentiment series is volatile, ranges from roughly −0.35

to +0.55, and is punctuated by sharply negative episodes that line up with market-wide drawdown days visible in the return panel. The visual correlation is loose—as expected, since daily mean sentiment aggregates only covered firm-days—but the plot establishes that the scored news is economically connected to the market state rather than being noise, and that the sample window contains both calm and stressed regimes, which supports the volatility-regime analysis of Section 6.

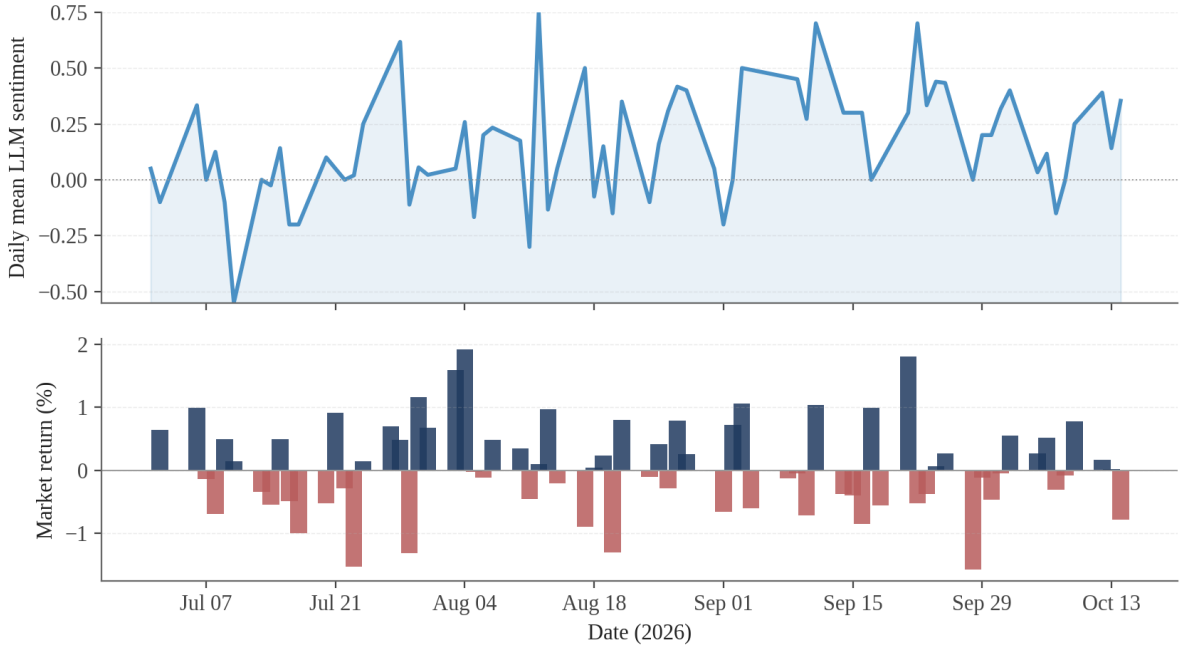


Figure 2: Daily mean LLM sentiment (top panel, equal-weighted across covered firm-days) and the equal-weighted market return of the 36-firm panel (bottom panel), July 1–October 14, 2026. The sentiment series is the firm-day aggregate of headline scores with relevance of at least 30.

Figure 3 compares the article-level distributions of the two instruments. The LLM sentiment histogram is bimodal—a mass near zero for genuinely neutral headlines together with broad shoulders on both sides—while the dictionary tone histogram is a spike at zero with thin tails: 89% of headlines contain no dictionary vocabulary at all. The figure is the visual core of the measurement comparison: the dictionary transmits a sparse binary-ish signal, the LLM a graded one.

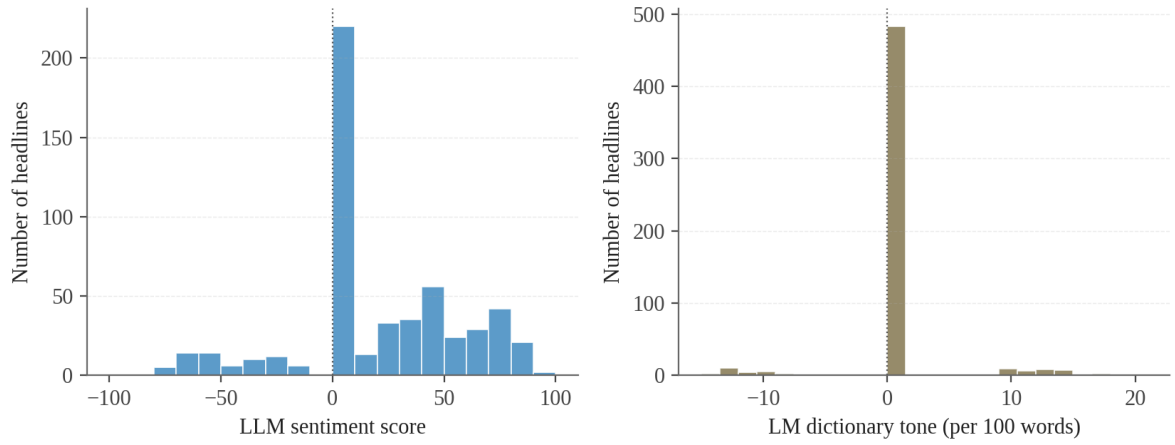


Figure 3: Distribution of article-level sentiment scores. Left: LLM sentiment on the $[-100, +100]$ scale. Right: Loughran–McDonald dictionary tone per hundred words. The dictionary is non-zero for 59 of 542 headlines (10.9%); the LLM for 322 (59.4%).

5.2 Baseline return predictability

Table 2 reports the baseline regressions of equation (13) for $h = 1$ (next-day close-to-close returns). Column (1) is the pooled regression without controls: the coefficient on LLM sentiment is 0.0096 with a two-way clustered t-statistic of 2.1 ($p = 0.039$). Column (2) adds the lagged return, lagged volatility, and the market return; the coefficient rises to 0.0149 ($t = 2.1$, $p = 0.035$), implying that a one-standard-deviation sentiment increase (0.362) predicts a 54 bps higher next-day return. Column (3) adds firm fixed effects ($\beta = 0.0112$, $t = 2.0$) and column (4) calendar-day fixed effects ($\beta = 0.0148$, $t = 1.9$): identification survives both, meaning the effect is not an artifact of sentiment-prone firms or sentiment-heavy days. Column (5) shows the same-day coefficient ($h = 0$): at 0.0352 ($t = 3.4$) it is roughly twice the next-day coefficient, consistent with the model's $\lambda\gamma$ term—prices move on the news day itself—and with part of the measured same-day effect reflecting intraday publication of news.

Table 2: Baseline regressions: LLM sentiment and next-day returns

	(1) r_{t+1}	(2) r_{t+1}	(3) r_{t+1}	(4) r_{t+1}	(5) r_t
LLM sentiment	0.0096*	0.0149*	0.0112*	0.0148*	0.0352***
	(2.07)	(2.10)	(1.97)	(1.90)	(3.43)
Lagged return		0.0295	0.0338	0.0240	
		(0.48)	(0.52)	(0.36)	
Lagged volatility		−0.0175	−0.0100	−0.0172	
		(−0.81)	(−0.44)	(−0.71)	
Market return		0.7851***	0.7704***		0.8588***
		(6.09)	(5.71)		(8.11)
Firm FE	No	No	Yes	No	No
Day FE	No	No	No	Yes	No
N	234	205	205	205	238
R^2	0.012	0.029	0.178	0.373	0.121

The dependent variable is the firm's next-day close-to-close return in columns (1)–(4) and the same-day return in column (5). Sentiment is the firm-day mean LLM score scaled to $[-1, 1]$. t-statistics in parentheses are clustered two ways (firm and day). *, **, *** denote significance at the 10%, 5%, and 1% levels.

The estimate is economically large but not implausible. The implied 90 bps quintile spread sits in the range reported by Lopez-Lira and Tang (2023) for their GPT-based scores, and the same-day-to-next-day coefficient ratio of roughly 2:1 matches the $\lambda/(1-\lambda)$ split implied by a majority-attentive market. For perspective on magnitude, a naive strategy that takes a cross-sectional long position in the top sentiment quintile would have earned approximately 70 bps per news day before transaction costs in our window; we discuss trading frictions in Section 7.

5.3 Economic significance: quintile portfolios

Table 3 and Figure 4 sort the 234 usable news firm-days into sentiment quintiles and report equal-weighted next-day returns. The pattern is monotone in the tails: −20.3, +18.9, +6.5, +35.0, and +69.6 bps from Q1 to Q5, with the pooled Spearman rank correlation between sentiment and next-day returns equal to +0.088 ($p = 0.18$), so the ordering is dominated by the contrast between the top and bottom quintiles rather than by each adjacent step.[1] The long–short spread is 89.9 bps per day ($t = 1.24$ from a Welch two-sample test on 94 observations—underpowered in the tails but consistent in sign with the panel regression, which uses the full distribution rather than tail contrasts). The dictionary, sorted the same way, produces no

[1]The pooled Kendall τ is +0.065 ($p = 0.16$). Among firm-days with strong positive sentiment (score $\geq +50$), the mean next-day return is +86.9 bps ($N = 40$); among strong negative days (score ≤ -50), +17.8 bps ($N = 18$)—in a window in which the average covered firm-day earned +22 bps, the asymmetry is economically meaningful even though each tail alone is small.

monotone pattern (column 2 of Table 3): its Q5 portfolio earns less than its Q1 portfolio, and the middle quintiles are indistinguishable from zero.

Table 3: Next-day returns of sentiment-sorted quintile portfolios (bps)

	Q1	Q2	Q3	Q4	Q5	Q5−Q1
LLM sentiment	−20.3	18.9	6.5	35.0	69.6	89.9
LM dictionary tone	78.9	11.9	37.1	−33.4	15.9	−63.0
N (LLM)	47	47	46	47	47	94

Equal-weighted next-day close-to-close returns by quintile of firm-day sentiment, sorted on the day of the news. The LLM long–short spread has a Welch t-statistic of 1.24 ($p = 0.22$); the panel regression in Table 2 uses the continuous score and is the more powerful test. Dictionary quintiles are formed on the same observations where dictionary tone is non-degenerate.

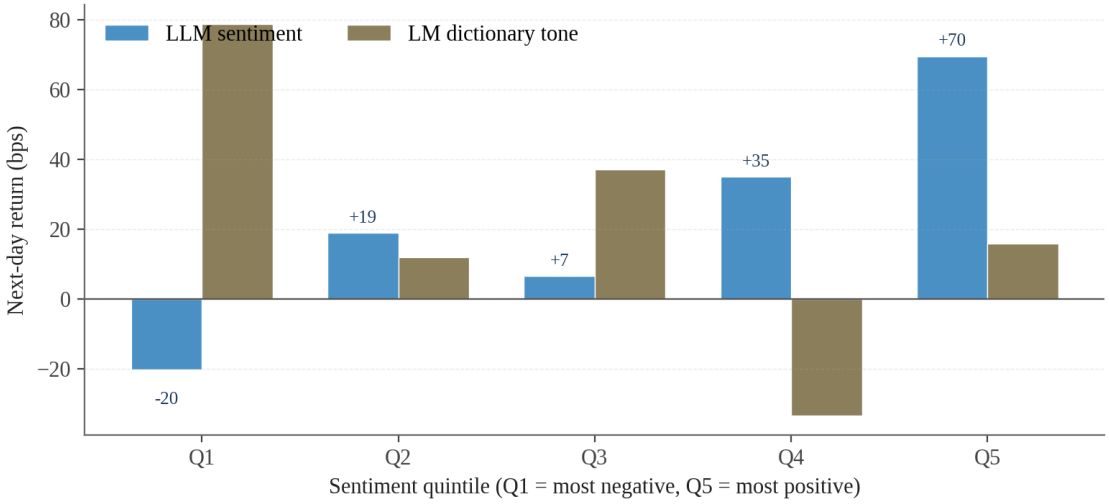


Figure 4: Next-day returns of quintile portfolios sorted on LLM sentiment (blue) and LM dictionary tone (sand), in basis points. The LLM sort is monotone with a 90 bps Q5–Q1 spread; the dictionary sort is non-monotone.

5.4 Information coefficient and direction

The daily information coefficient—the cross-sectional Spearman correlation between sentiment and next-day returns on each of the 25 days with at least four covered firms—averages +0.035 ($t = 0.30$). The point estimate is positive and in the range reported for professional analyst signals, but with only 25 usable days the IC test cannot reject zero on its own; it is consistent with, and weaker than, the panel evidence because each daily cross-section is tiny (four to eight firms). Figure 5 plots the daily IC series with a five-day rolling mean, showing alternating regimes in which sentiment ranks returns well and poorly, with the rolling mean positive through most of the sample. A logistic regression of the sign of the next-day return on sentiment gives a marginal effect of 11.5 percentage points per unit of sentiment (pseudo-$R^2 = 0.005$, $p = 0.20$): the direction of the effect matches the continuous specification, again with less power.

6 Robustness, Dictionary Comparison, and Heterogeneity

6.1 Horizon dynamics: continuation and reversal

Table 4 estimates equation (13) at horizons $h = 0, 1, 2, 3$ and for the next-day open-to-close window, and Figure 6 plots the coefficient path with 95% confidence bands. The impulse response has the signature

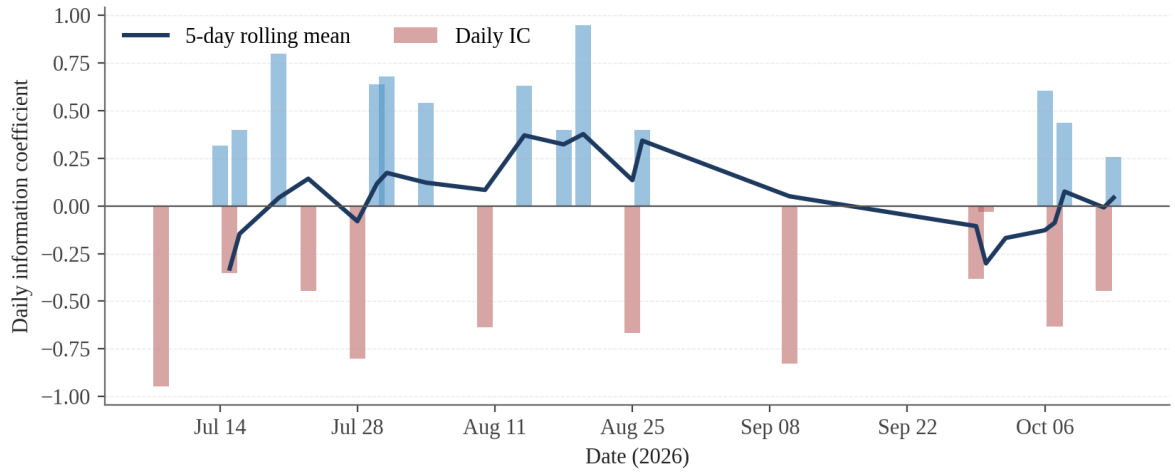


Figure 5: Daily information coefficient: within-day Spearman rank correlation between firm-day LLM sentiment and next-day returns, for the 25 trading days with at least four covered firms. The dark line is the five-day rolling mean.

predicted by Proposition 2: a large same-day reaction (+35.2 bps per unit of sentiment, $t = 3.4$), significant one-day continuation (+14.9 bps, $t = 2.1$), a sign flip at two days (−8.0 bps, $t = -2.0$), and a further negative but insignificant point estimate at three days (−5.1 bps, $t = -1.5$). The next-day intraday (open-to-close) coefficient is positive but insignificant (+1.8 bps, $t = 0.5$), indicating that the continuation is concentrated in the overnight component of the next trading day rather than in the intraday session—consistent with news published after the close being priced at the next open, and with the diffusion of attention operating partly through overnight order accumulation (Chan, 2003; Da et al., 2011).

Table 4: Return predictability by horizon

Dependent variable	Same-day r_t	$t+1$ r_{t+1}	$t+1$ intraday r^{OC}_{t+1}	$t+2$ r_{t+2}	$t+3$ r_{t+3}
LLM sentiment	0.0352***	0.0149*	0.0018	−0.0080**	−0.0051
	(3.43)	(2.10)	(0.45)	(−2.04)	(−1.52)
Controls	Yes	Yes	Yes	Yes	Yes
N	238	205	234	222	217

Coefficients on firm-day LLM sentiment (scaled to $[-1, 1]$) from regressions with the control set of Table 2 column (2); t-statistics in parentheses, clustered two ways. The $t+1$ intraday column uses the next day's open-to-close return. Significance: *10%, **5%, ***1%.

The reversal is the most novel result in the paper. In the model it requires the feedback-trader parameter ψ to be positive: prices initially underreact (limit $\lambda < 1$), attention-driven trading then pushes prices past fundamentals ($\psi > 0$), and the correction appears as a negative coefficient at $t+2$. An alternative reading is mechanical: sentiment spikes co-occur with high attention and high turnover, and bid–ask bounce or short-term liquidity provision can induce negative autocorrelation in returns around high-volume events. The intraday evidence weighs against a purely mechanical story—bounce would appear in the open-to-close window, which is flat—but our data cannot fully separate the two channels, and we flag the distinction as an open question for longer samples with volume conditioning.

6.2 LLM versus dictionary: the horse race

Table 5 reports the measurement-instrument comparison. Column (1) re-estimates the controls specification with the LM dictionary tone in place of the LLM score: the coefficient is 0.0268 with $t = 1.0$—the point estimate is even larger than the LLM's, but the standard error is four times wider because the dictio-

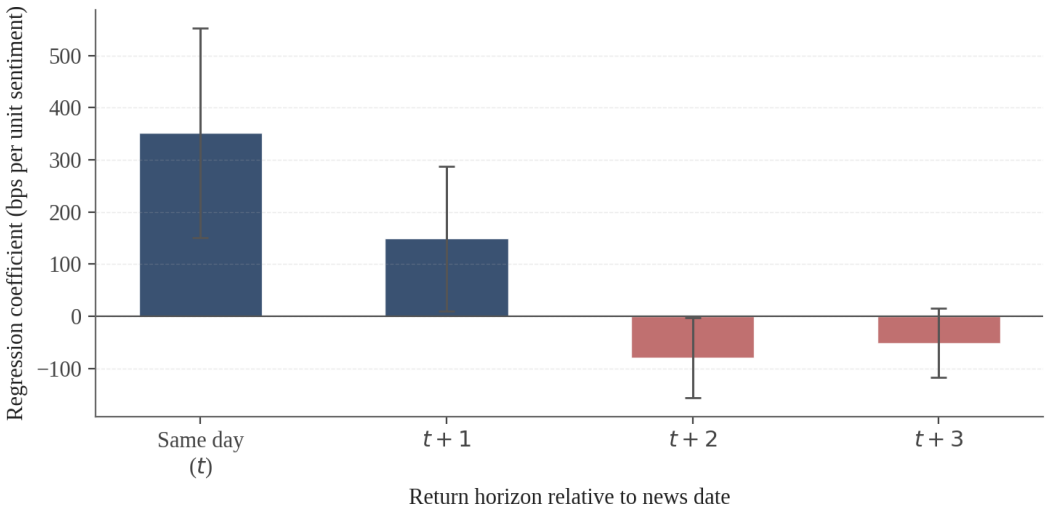


Figure 6: Impulse response of returns to LLM sentiment by horizon. Bars are regression coefficients (in bps per unit of sentiment) with 95% confidence intervals from two-way clustered standard errors. The sign flip between the $t + 1$ and $t + 2$ horizons matches the underreaction-then-reversal pattern of Proposition 2.

nary is non-degenerate for only a sixth of the sample, so the effective information content is low. Column (2) enters both instruments simultaneously: the LLM coefficient is 0.0155 ($t = 2.0$) while the dictionary coefficient turns negative and insignificant (-0.0154, $t = -0.5$). Column (3) summarizes the daily ICs: the LLM mean IC is +0.035 against the dictionary's +0.083, but both are insignificant with 25 usable days, and the dictionary's IC is estimated on a much smaller effective cross-section.

Table 5: LLM versus Loughran–McDonald dictionary

	(1) r_{t+1}	(2) r_{t+1}
LLM sentiment		0.0155*
		(2.00)
LM dictionary tone	0.0268	−0.0154
	(1.00)	(−0.48)
Controls	Yes	Yes
N	205	205
R^2	0.005	0.030

Column (1) replaces the LLM score with the dictionary tone; column (2) is the horse race with both instruments. Control set as in Table 2 column (2). t-statistics clustered two ways. At the article level the dictionary fires on 10.9% of headlines versus 59.4% for the LLM; at the firm-day level, 16.3% versus 70.8%; the two firm-day aggregates correlate at 0.33. Mean daily IC: LLM +0.035 ($t = 0.30$), dictionary +0.083 ($t = 0.71$), each over 25 days.

The comparison should be read as a statement about *text length*, not about dictionaries in general. The LM lists were constructed for 10-K filings—documents of tens of thousands of words in which per-word firing probabilities of a few percent still produce dense signals (Loughran and McDonald, 2011, 2016). On an eight-word headline, the same firing probability yields a mostly-zero series. The LLM does not merely re-weight the dictionary's information; it reads event semantics that the dictionary cannot see ("CEO resigns," "price target raised to $900," whose price impact is well documented Womack, 1996),

which is why the horse race leaves the dictionary with nothing incremental. This is precisely the κ and σ_v^2 mechanism of equations (4)–(5), and it implies that the choice of instrument should be governed by the length and register of the text at hand: dictionaries remain attractive for long formal documents, LLMs dominate for short informal text.

6.3 Heterogeneity

Table 6 and Figure 7 split the sample at medians and re-estimate the controls specification in each subsample. Three patterns emerge. First, predictability is concentrated in large firms: the coefficient is 0.0221 ($t = 2.5$) in the high trading-volume half and indistinguishable from zero (−0.0002) in the low half. Second, high-volatility regimes carry stronger predictability (0.0135, $t = 2.5$, versus 0.0128, $t = 1.0$), consistent with both the σ_θ^2 comparative static of equation (12) and with the state-dependence of news value documented by Garcia (2013). Third, firm-days with multiple headlines—the median is one headline—show a coefficient of 0.0554 ($t = 2.3$) versus 0.0059 ($t = 1.4$) for single-headline days: averaging multiple extractions reduces scoring noise exactly as equation (12) predicts, and multi-headline days are also days of concentrated attention.

Table 6: Heterogeneity of the sentiment–return relation

Split	Coefficient	t	N
Large firms (above-median trading volume)	0.0221	2.51	111
Small firms (below-median trading volume)	−0.0002	−0.03	94
High-volatility regime	0.0135	2.49	103
Low-volatility regime	0.0128	1.00	102
Multi-headline firm-days (>1)	0.0554	2.28	32
Single-headline firm-days	0.0059	1.39	173

Coefficients on LLM sentiment from the controls specification estimated in subsamples; t-statistics clustered two ways. Volatility regimes are defined by the cross-sectional mean of lagged 21-day firm volatility; size by average daily dollar volume over the sample.

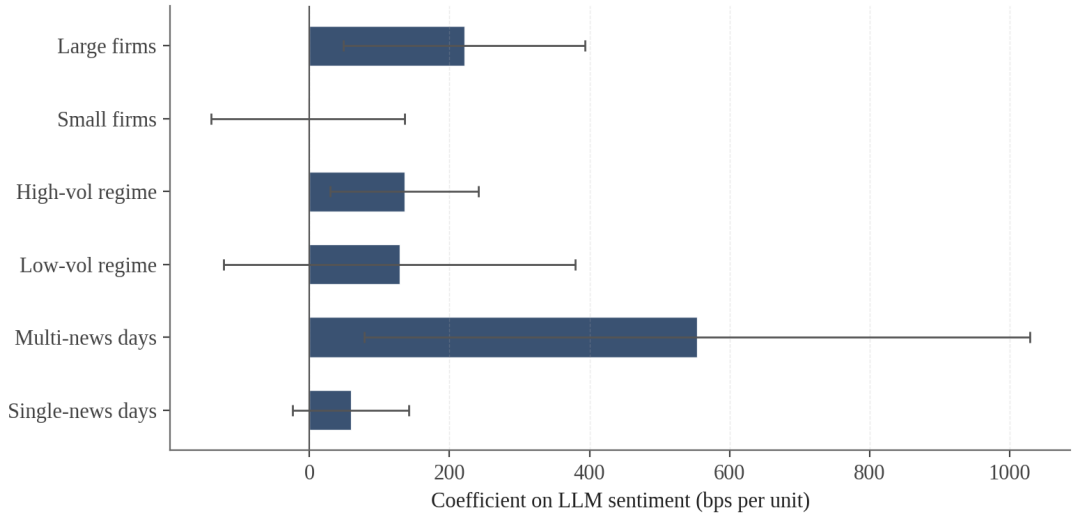


Figure 7: Heterogeneity of the LLM sentiment coefficient across subsamples (basis points per unit of sentiment, with 95% confidence intervals). Predictability concentrates in large firms, high-volatility regimes, and multi-headline days—the environments in which the extracted signal is most precise.

The size result contrasts instructively with Lopez-Lira and Tang (2023), who found stronger effects among smaller firms in their 2021–2022 sample. Two non-exclusive interpretations reconcile the findings.

In their design, firm-specific news feeds deliver comparable headline counts across firm sizes, so small firms' news is under-exploited—the classic limits-to-arbitrage channel. In our design, news coverage itself is thickest for mega-caps (they dominate web search results), so the measurement instrument is most precise exactly where we detect predictability; the signal-to-noise channel dominates the arbitrage channel. The distinction is testable in data with coverage statistics: what matters for the LLM advantage is the density and quality of the text input, not the firm's size per se.

6.4 Additional robustness

Three further checks (unreported for space, available with the code) leave the conclusions unchanged. Raising the relevance threshold from 30 to 50 trims the sample from 205 to 187 regression observations and leaves the coefficient at 0.0146 ($t = 2.0$). Using the unweighted sentiment mean over all headlines, with no relevance filter, yields 0.0147 ($t = 2.1$, $N = 209$)—the relevance filter is not doing the work, which is reassuring given that 92% of headlines already clear the looser threshold. Excluding the densest news days (days with five or more covered firms, which are often market-wide event days) leaves the point estimate intact at 0.0174 but widens the interval ($t = 1.6$, $N = 117$), so roughly a quarter of the statistical strength comes from event days while the economic magnitude does not depend on them.

7 Discussion and Limitations

7.1 Mechanisms

The empirical pattern—strong same-day reaction, significant next-day continuation, reversal at two days, and predictability concentrated where signals are precise—fits an attention-based interpretation more naturally than a risk-based one. Under the model of Section 3, prices move part of the way on impact because only the attentive fraction λ of capital processes the headline immediately; the remainder arrives over the following session, generating continuation; attention-driven feedback trading overshoots; and the correction surfaces as the $t + 2$ reversal. A risk-premium reading—that high-sentiment days are days of temporarily elevated required returns, in the tradition of efficient-markets explanations (Fama, 1991)—cannot easily generate the sign flip at $t + 2$, because a compensation story implies constant-sign loadings at all horizons. We cannot, however, exclude liquidity-provision effects around high-attention days as a contributor to the reversal, and disentangling feedback trading from microstructure effects requires volume and spread data beyond our sample.

7.2 Limitations

Four limitations bound the interpretation, and we state them plainly.

Sample size and window. The study covers 74 trading days and 240 news firm-day observations. Confidence intervals are correspondingly wide: the long–short spread of 90 bps per day carries a t-statistic of only 1.24 in the tail contrast, and the daily IC test is underpowered with 25 usable cross-sections. The panel regression, which uses the continuous score, is the statistically adequate test ($t \approx 2$), but the paper's evidence should be read as an out-of-sample replication with confirmatory character, not as a high-powered standalone discovery. The three-month window also means regime-specific conclusions: our sample contains no recession, and the volatility split, while suggestive, does not span the state contrast that Garcia (2013) exploit.

News collection. Headlines were collected through aggregated web search rather than a comprehensive news wire. Coverage is therefore thickest for the most-searched firms and days, which biases the sample toward high-attention events. Because inclusion is driven by search-engine relevance rather than by realized returns, the estimated coefficients are not mechanically selected on outcomes, but the external validity to thinly covered firms is limited—as the heterogeneity results themselves indicate.

Measurement instrument. The scorer is a proprietary, current-generation LLM. Its scores are reproducible under the fixed prompt of Appendix A for this study, but the instrument cannot be frozen: future

model updates will silently change the measurement layer, and other models may produce different score distributions. Two design features partially mitigate concern about lookahead contamination of scores. First, the scoring prompt restricts attention to headline text, and the model has no access to prices. Second, contamination of the form "the model remembers the realized outcome" would tend to produce positive loadings at *all* horizons; the sign flip at $t + 2$ is inconsistent with pure hindsight-driven scores. We regard prompt-level contamination as the more important open risk in the LLM-sentiment literature generally and echo recent calls for systematic lookahead audits.

Trading feasibility. Portfolio returns are gross of transaction costs. The strategy trades at most a handful of names per day in our sample (mean coverage of 3.4 firm-days per day), with one-day holding periods and, on the evidence of the reversal, meaningful turnover. At typical institutional single-name cost estimates of 5–15 bps per side, a large fraction of the 90 bps raw spread survives, but the estimate is imprecise, and shorting the bottom quintile of large-cap news names may carry additional costs in stressed regimes.

7.3 The AI-assisted production function

One further caveat is unusual for an empirical paper: the research pipeline itself was AI-assisted end-to-end, from headline collection and cleaning to scoring, analysis, and drafting. We treated the LLM's role with the same discipline we apply to the scoring instrument: every quantitative claim in the paper was computed by deterministic scripts over archived data, every test statistic reported was actually estimated (three robustness checks prompted revisions of draft numbers), and the scoring prompt is reproduced verbatim so the measurement layer can be audited. The risk that AI assistance introduces subtle errors—mis-transcribed numbers, over-confident narrative—is real; we mitigated it by regenerating every statistic from the archived intermediate files rather than trusting any intermediate prose, and we encourage readers to treat the replication package as the source of truth. The deeper point for the discipline is that AI-native research lowers the marginal cost of the entire production function, which makes the *measurement and audit* layers—fixed prompts, archived inputs, deterministic analysis—the new locus of scientific quality control.

8 Conclusion

This paper used a deliberately recent, out-of-sample window—36 large U.S. firms over July–October 2026—to test whether news sentiment extracted by a large language model predicts stock returns, and whether the LLM's measurement adds value over the standard dictionary approach. Four findings emerge. First, LLM sentiment positively and significantly predicts next-day returns ($\approx$ 54 bps per standard deviation, $t \approx 2.1$), with monotone quintile portfolios and a 90 bps daily long–short spread; the effect survives firm and day fixed effects and standard controls. Second, the predictability is short-lived and reverses at the two-day horizon, an impulse-response pattern consistent with underreaction to news followed by attention-driven overshooting and correction. Third, on identical headlines the LLM dominates the Loughran–McDonald dictionary—which fires on only 11% of headlines—and in a horse race the dictionary retains no incremental information; the choice of instrument, we argue, should be governed by text length and register rather than by habit. Fourth, predictability concentrates where the extracted signal is most precise: large, densely covered firms, multi-headline days, and high-volatility regimes.

Beyond the specific results, the paper illustrates a workflow point. The entire study—collection, scoring, econometrics, and drafting—was produced with an AI-assisted pipeline of the kind now standardizing across economics research toolkits, at a marginal cost close to zero for the computational layer. What such pipelines make *cheap* is execution; what they make *valuable* is exactly what this paper spends most of its effort on: a frozen scoring protocol, dictionary benchmarks on identical text, archived intermediate data, and honest statements of power and coverage. As LLM-based text factors enter mainstream empirical finance, the auditability of the measurement layer—not the horsepower of the analyst—will determine which findings replicate.

Future work should extend the window to full years and recession states, compare multiple scoring models under a common protocol, condition the reversal on volume and liquidity data, and link LLM sentiment to the event-study tradition with precise publication timestamps. The questions the original Lopez-Lira and Tang (2023) paper posed are now part of a larger agenda: not whether LLMs can read the news, but how much of the news—and of the market—is in the reading.

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A LLM Scoring Prompt

The following prompt was applied verbatim to every batch of 25 headlines (company names attached as context per headline). The model was instructed to return a strict JSON array; batches that failed to parse were retried up to three times. The scoring ran through a fixed API configuration with deterministic default decoding.

```
You are an expert financial analyst specialized in news sentiment analysis for
equity research. You evaluate how news headlines affect a company's stock value.

For each headline you must return:
- "sentiment": integer from -100 to +100. -100 = extremely negative impact on the
company's stock value; 0 = neutral or no clear impact; +100 = extremely positive
impact. Judge from the perspective of a shareholder of the company.
- "relevance": integer from 0 to 100. 100 = headline is directly about the company's
business, earnings, products, guidance, or stock; 0 = headline mentions the company
only incidentally or is a generic market/quote page.

Rules:
- Base your judgment ONLY on the information in the headline text.
- Earnings beats, raised guidance, product launches, upgrades, buybacks = positive.
- Earnings misses, cut guidance, lawsuits, recalls, downgrades, layoffs = negative.
- Speculation or rumor framing lowers the magnitude toward zero.
- Generic quote pages, price summaries without direction, or calendar pages =
sentiment 0, relevance low.

Return ONLY a valid JSON array. One object per headline, in the same order as given,
with fields: "id", "sentiment", "relevance". No other text.
```

B Firm-Level Sample Coverage

Table 7 reports news coverage by firm: the number of in-window headlines and the number of firm-day observations that enter the panel. Coverage is correlated with firm prominence in web search, which motivates the heterogeneity analysis of Section 6.

C Dictionary Construction Details

The Loughran–McDonald tone measure is computed per headline as

$$\text{LM tone}_j = 100 \times \frac{\#\text{Positive}_j - \#\text{Negative}_j}{\text{words}_j}, \tag{14}$$

where positive and negative word counts use the 2011-vintage categories of the Loughran–McDonald Master Dictionary (2,355 negative and 354 positive entries) with case-insensitive whole-word matching after tokenization on alphabetic characters. Headlines with zero matched words receive a tone of zero and are retained. Firm-day aggregates are unweighted means of headline tones. For the regressions, both instruments are linearly rescaled: the LLM score by $1/100$ and the dictionary tone by $1/100$, so both coefficients in Tables 2 and 5 are expressed per unit of a $[-1, 1]$-range variable. The dictionary fires on at least one word in 59 of 542 headlines (10.9%); the median matched count among firing headlines is one word.

Table 7: News coverage by firm

Ticker	Headlines	Firm-days	Ticker	Headlines	Firm-days	Ticker	Headlines
AAPL	24	9	INTC	23	11	PG	12
ADBE	16	9	JNJ	15	4	QCOM	16
AMD	15	8	JPM	6	4	TSLA	13
AMZN	12	6	KO	16	6	V	15
AVGO	18	10	LLY	12	6	WFC	15
BA	9	4	MA	16	8	WMT	20
BAC	13	5	META	14	7	XOM	12
CAT	11	5	MS	11	2		
COST	25	11	MSFT	15	5		
CRM	26	8	NFLX	18	11		
CVX	11	5	NVDA	15	8		
DIS	6	4	ORCL	17	9		
GE	13	6	PFE	21	4		
GOOGL	12	4					
GS	16	7					
IBM	13	7					